Jorge Julián Acero Lomelí

Senior Software Engineer
hi@imjulian.com | linkedin.com/in/imjulian | imjulian.com | +524495950265 | Aguascalientes, México.

I'm a results-driven Senior Software Engineer with **8 years of experience** architecting highly scalable, real-time systems, including automated earthquake alerts supporting **2 million users** and digital surgical platforms for **Johnson & Johnson** by using AWS, Node.js, React, Go, Python, and building **AI Agents**.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | EPAM Systems | Full Time | Mar 2026 - Present
Remote | Aguascalientes, México
Johnson & Johnson

- Add support for bronchoscopy and urology procedures on Johnson & Johnson's Polyphonic platform.
- Implement a Kafka event-processing microservice to manage videos across multiple procedures.
- **Create a 3D model viewer** with VTK.js that reads VTP files, allowing doctors to visualize a patient's lungs and the planned surgical path with its target.
- Stack: Next.js, Nest.js, Node.js, React, TypeORM, PostgreSQL, Kafka, Kiro, and AWS.

Full Stack Software Engineer | SkyAlert | Full Time | Nov 2021 - Mar 2026
Remote | Aguascalientes, México

- Create a system that automatically generates earthquake reports in seconds, which sends valuable data for the alerts team and the public. This system successfully handles important earthquakes in México and publishes the data to the public through **the mobile app which has 2 million users** (eg. Sep 19, 2022, Jan 12, 2025, and Jan 2, 2026). Stack: AWS Lambda, Node.js, TypeScript, React, React Query, Tailwind CSS, SNS, SQS, Kinesis, API Gateway, and DynamoDB.
- **Cut database costs by up to 90%** by migrating from AWS Aurora to DynamoDB.
- Build an AI agent to troubleshoot IoT devices connectivity problems. Built with Python, strands, and AWS Bedrock.
- Implement new sensors with nvidia Jetson and AWS Greengrass which detect earthquakes using local AI models and Python.
- Use IaC tools like AWS CDK and SST to define infra and deploy every project described here to AWS.
- Build a system that is able to detect **microquakes** in real-time located in CDMX, this is the first system in the world to achieve this. This system is built with Golang, AWS Fargate, MQTT, IoT Core, Cloudwatch Alarms, SNS, Lambda URLs, SQLite, and DynamoDB.
- Maintain legacy systems working and connect them with the new ones through web scraping using Playwright.
- Maintain the main system and algorithm of **earthquake confirmation** which is built with TypeScript and JavaScript by using Kiro, TDD, and CI/CD pipelines.
- Improved sensor network connectivity from 70% to 98% stable through the development of internal tooling.
- Involved in the app's rating improvement from 3.8 to 4.6 stars on the App Store and Google Play.
- Build a weather dashboard using data provided by rain sensors and download NOAA data to generate forecast and satellite maps. Stack: Grafana, AWS Lambda, Node.js, TypeScript, Nest.js, React, and PostgreSQL.

Teacher Assistant | Código con Juan | Full Time | Jul 2020 - Nov 2021
Remote | Aguascalientes, México

- Provided rapid responses to student inquiries on Udemy, documenting ideas for future courses.
- Be aware of the breaking changes in libraries, frameworks, and packages that are used in the courses, which are about TypeScript, React, React Native, Tailwind CSS, Node, GraphQL, MongoDB, Laravel, and Cypress.

Full Stack Software Engineer | MKF Merkamorfosis | Contractor | Feb 2021 - Jul 2021
Hybrid | Aguascalientes, México

- Developed dynamic organization charts, which is built with Next.js, TypeScript, Firebase, Material UI and D3.js.
- Created a web app to manage political campaigns, which is built with Next.js, TypeScript, Firebase, and MUI.
- Enhanced communication within MKF by organizing employees into departments, reducing response times.

Full Stack Software Engineer | MANCOMM | Contractor | Oct 2020 - Dec 2020
Remote | Aguascalientes, México

- Replace SendOwl with Stripe for payment processing.
- Stack: Next.js, Node.js, Material UI, Stripe, C#, and SQL Server.

Full Stack Software Engineer | DCM AGUASCALIENTES | Contractor | Nov 2019 - Dec 2019
Remote | Aguascalientes, México

- Developed a project management web application with role-based access control (RBAC) and real-time data. Stack: MongoDB, Express.js, React.js, Node.js, GraphQL and Heroku.
- Improve the company's project management system by saving historical data about the deadlines and materials.

Power BI Data Analyst & Support | Softtek | Full Time | Jul 2019 - Jul 2020
Hybrid | Aguascalientes, México

- Analyzed sales data for Burger King, Tim Hortons and Popeyes by using Power BI.
- Managed custom modules for SAP.

Full Stack Software Engineer | RAEE Company | Contractor | Jan 2019 - Apr 2019
Hybrid | Aguascalientes, México

- Create a web app to control the access to the company's warehouse.
- Improve the company's access control to the warehouse by registering the employees and trucks who went inside the warehouse.
- Stack: MongoDB, Node.js, React.js, Express.js, REST and Heroku.

EDUCATION
Bachelor's Degree in Systems Engineering | Universidad Politécnica de Aguascalientes | Sep 2016 – Apr 2020

LANGUAGES
English: B2 | **Spanish:** Native

SKILLS
Languages: TypeScript, JavaScript, Python, Go (Golang), SQL, C#.
Frontend: React, Next.js, React Query, Tailwind CSS, Material UI (MUI), D3.js, VTK.js.
Backend: Node.js, NestJS, TypeORM, REST APIs, Microservices, Event-Driven Architecture, Design Systems, Apache Kafka, MQTT, Stripe.
Cloud (AWS): Lambda, API Gateway, SNS, SQS, Kinesis, DynamoDB, Aurora, Fargate, IoT Core, IoT Greengrass, Bedrock, CloudWatch.
Databases: PostgreSQL, DynamoDB, Amazon Aurora, SQLite, Firebase, SQL Server, MongoDB.
DevOps & Testing: Infrastructure as Code (AWS CDK, SST), CI/CD, Test-Driven Development (TDD), Playwright, Grafana.
AI & IoT: AI Agents (Strands Agents, AWS Bedrock), Edge AI on NVIDIA Jetson, IoT Sensor Networks, Kiro, Claude.